

SOHAM BHATTACHARYA

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PROJECTS

- **Multimodal Sentiment Analysis** *Jun.'25 - present, ISI Kolkata*
Novel perspective on multimodal sentiment analysis through underutilized yet emotionally rich cues in real-world media
Keywords & Models: Ensemble Model Fusion, Sentiment Analysis, DistilRoBERTa, BLIP, CLIP
- **K8sCIFAR** *Jan.'25 - Apr.'25, RKMVERI*
Demonstrated distributed data parallelism on CIFAR-10 using Kubernetes
Tools: PyTorch, Kubernetes, Docker
- **Raag Sense** *Jan.'25 - Apr.'25, RKMVERI*
A pioneering DL-based system for raga recognition in Indian Classical Music (ICM), setting a new benchmark
Keywords: Data curation, ICM-aware feature extraction, MFCC, CNN
- **ChessLens** *Jan.'25 - Apr.'25, RKMVERI*
A cost-effective, vision-based system for accurate chess piece tracking and board digitization, poised to be a potential alternative to DGT chessboards
Keywords: open-cv, occupancy detector
- **Rhythms of the Mind** *Aug.'24 - Nov.'24, RKMVERI*
An intelligent model that analyzes music consumption patterns to detect changes in mental health, supporting personalized music therapy
Keywords: Python, Kaggle, EDA, feature engineering, fine-tuning
- **Savoring Sentiments** *Jan.'24 - Apr.'24, IIT Madras*
An ML model that predicts restaurant's recipe ratings by analyzing user reviews and key engagement metrics, combining textual sentiment with numerical insights.
Keywords: EDA, TF-IDF Vectorizer, LightGBM

COURSEWORK

RKMVERI Belur:

Deep Learning • Natural Language Processing • Computer Vision
• Artificial Intelligence • Time Series Forecasting • Survival Analysis
• Machine Learning • Statistics • Java & Hadoop • Econometrics
• Finance • Reinforcement Learning • DL in Cyber Security

Indian Institute of Technology Madras:

Programming in Python • Machine Learning • Tools in Data Science
• Business Data Management • Business Analytics • Statistics

ACCOMPLISHMENTS

- **Badges earned:** Introduction to **GenAI** (Google Cloud), **Machine Learning Foundations** (AWS Educate) *(Aug.'25)*
- **Udacity:** Introducing **Generative AI** with AWS *(Aug.'25)*
- Attended **Crest Crypto Summer School 2025**, TCG Crest *(June 2025)*
- NPTEL Workshop: Time Series Forecasting using **LSTM and ARIMA** *(Jun.'2025)*
- **Topper** in courses: Statistics, Data Structures & Algorithms (**RKMVERI**)
Tools in Data Science, Programming in Python (**IIT Madras**)
- **Coursera:** Automation & Advanced Techniques with **Copilot in Excel** by Microsoft *(Jul.'25)*, Introduction to **Machine Learning on AWS** *(Feb.'25)*, Google Crash Course in Python *(Sept.'22)*
- **NPTEL:** Competitive Programming *(Nov.'24)*
- **Google:** **Google Analytics** Individual Qualification *(Jul.'24)*

EXPERIENCE

Research Intern

Jun.'25 - present ISI, Kolkata

Advisor- Dr. Umapada Pal & Dr. Ali Reza Alaei

EDUCATION

- **Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI), Belur, Howrah**
M.Sc. in Big Data Analytics
(Rank: 5/27)
2024 - Present (Sem-2) CGPA: 7.83
- **Indian Institute of Technology Madras**
Diploma in Data Science
2021 - 2024 CGPA: 8.54
- **St. Xavier's College Kolkata**
B.Sc.(H) in Mathematics
2021 - 2024 CGPA: 6.72
- **Don Bosco School Liluah**
Indian School Certificate (10+2)
2019 - 2021 Score: 94.25%
Indian Certificate for Secondary Education (10)
2008 - 2019 Score: 90%

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL, $\text{\LaTeX}$, HTML, CSS
- **Frameworks:** PySpark, Hadoop, Pregel
- **Databases:** MySQL, Neo4j
- **Tools:** R, Docker, Git/Github, MS Excel, Google Analytics, PowerBI, Looker Studio, Flourish, Kumu

LEADERSHIP POSITIONS

- **RKMVERI**
 - Placement Coordinator *[May'25 - Present]*
 - Organizing Committee Member, Percepton 2025 (annual departmental tech fest) *[Jan.'25]*
 - Class Representative *[Jul.'24 - Present]*
- **IIT Madras**
 - Group Leader, Student Council IITM BS *[Aug.'23 - Aug.'24]*